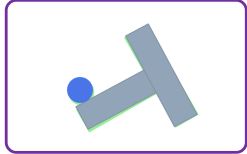
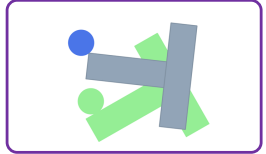
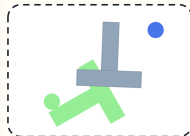


Context (PushT)

Goal state



Two nearby candidate actions



Predicted future (plausible)



Candidate A (fail)



Predicted future (plausible)



Candidate B (success)

Observed Mismatch

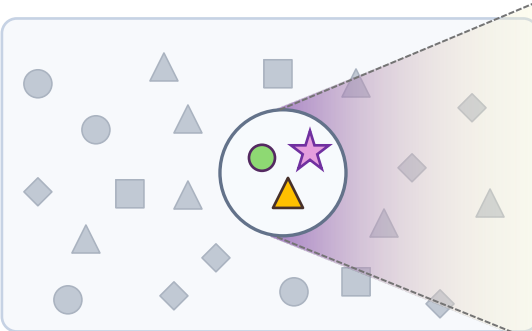
D-JEPA learns decision relations and writes them back into the latent space.

Decision Geometry Alignment

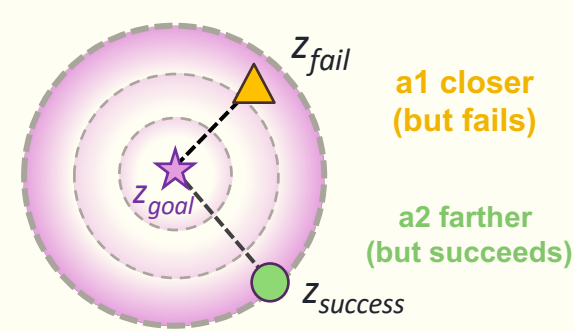
Closer in prediction, worse in action.

$$d_{pred}(z_{fail}, z_{goal}) < d_{pred}(z_{success}, z_{goal})$$

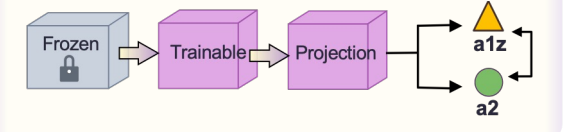
Native predictive latent geometry



Decision neighborhood



Relational Learning



Geometry Write-back



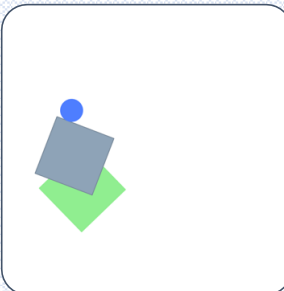
Native Deployment (PushT confirmation)

|         |             |           |
|---------|-------------|-----------|
| D-JEPA  | ●●●●●●●●●●○ | 225 / 256 |
| LeWM    | ●●●●●●●●●○  | 214 / 256 |
| TD-JEPA | ●●●●●●●●○   | 197 / 256 |

Object Manipulation



PushT

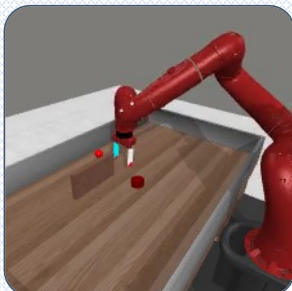


PushObj

Reaching Control

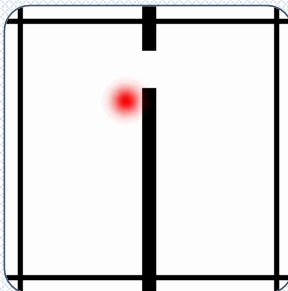


Reacher



Reach-Wall

Goal-Conditioned



Two-Room



PointMaze

Arm Manipulation



RoboTwin cooperative lifting

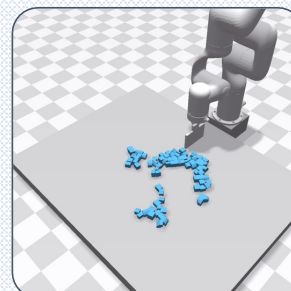


Dual-object manipulation

Granular Shaping



Cube

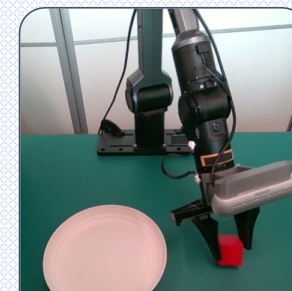


Granular

Real-world manipulation



Take pen

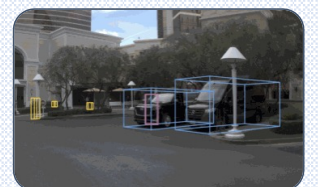


Stack cube

Autonomous Driving



Navigating



Driving